MARINDUQUE STATE UNIVERSITY

College of Information and Computing Sciences

Bachelor of Science in Information Technology

DELIVERABLE #7 — CAPSTONE PROJECT

G-TRAMS

Gasan Tricycle Regulatory and Automated Management System

A Comprehensive Web-Based Management and Enforcement System for Motorized Tricycle Operator's Permits (MTOP), Franchise Compliance, and Revenue Remittance in the Municipality of Gasan, Marinduque.

USER'S MANUAL OF THE SYSTEM

SYSTEM DEVELOPMENT TEAM (GROUP 5)

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Official System Version: 2.4.0 (Production Release)
Deployment Environment: Vercel Global Edge & Cloudflare Worker API
Academic Term: Academic Year 2026–2027 • Municipality of Gasan, Marinduque

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Notice for System Evaluators & Users

This User's Manual is written for end-users operating the actual working system. In accordance with Marinduque State University CICS guidelines, technical programming code, database schemas, and inner algorithmic implementations are omitted in favor of operational workflows, verified procedures, and authentic system screenshots.

1. SYSTEM OVERVIEW

1.1 System Description

G-TRAMS (Gasan Tricycle Regulatory and Automated Management System) is an enterprise-grade, web-based regulatory management and enforcement platform engineered specifically for the **Municipality of Gasan, Marinduque**. The system serves as the centralized digital infrastructure for managing Motorized Tricycle Operator's Permits (MTOP), organizing Tricycle Operators and Drivers Association (TODA) member rosters, streamlining municipal treasury fee collections, enforcing local traffic ordinances via mobile QR verification, and ensuring full regulatory transparency across all public transport units operating within municipal boundaries.



Figure 1. Official G-TRAMS Public Landing Portal (https://g-trams-official.vercel.app/)

1.2 Purpose and Objectives

The primary purpose of G-TRAMS is to replace obsolete, paper-heavy filing cabinets and slow manual processing with an integrated, automated, and secure digital environment. Specific objectives include:

- **Eliminate Application Delays:** Provide operators with a 24/7 online application portal featuring automated document scanning and optical verification, reducing turnaround times from several weeks to just a few days.
- **Prevent Fraud & Duplicate Franchises:** Enforce strict, automated body number allocation and duplicate prevention algorithms, eliminating unauthorized "colorum" units and body number conflicts.
- **Ensure Real-Time Field Verification:** Empower traffic enforcers and police officers with instant QR code scanning on mobile devices to verify vehicle legality, route zone authorization, and driver validity on the road without physical paperwork.
- **Streamline Municipal Treasury Remittance:** Provide a dedicated Cashier Terminal for issuing official receipts, tracking cash and digital GCash payments, and generating daily COA-compliant revenue remittance sheets.
- **Comply with Data Privacy Laws:** Protect personal sensitive data of operators and drivers in strict compliance with the Data Privacy Act of 2012 (Republic Act No. 10173).

1.3 Intended Users and Roles

USER ROLE	PRIMARY RESPONSIBILITY	SYSTEM ACCESS PORTAL
Citizen / Tricycle Operator	Registers accounts, applies for new/renewal franchises, uploads vehicle OR/CR and credentials, tracks permit progress, and downloads digital MTOP credentials.	Operator Portal & Mobile Web Garage
TODA President	Maintains association rosters, registers individual members, issues TODA clearances, and submits formal member removal requests.	TODA Management Console
Traffic Enforcement Officer	Performs on-road QR code scanning of operator IDs and vehicle stickers, verifies compliance, and issues traffic citation tickets.	Enforcer Mobile Web Scanner
Municipal Cashier / Treasury	Assesses regulatory fees, accepts cash or GCash payments, logs official receipts (OR), and prints daily revenue remittance sheets.	Cashier & Payment Terminal
BPLO Franchising Officer	Inspects submitted documents, verifies AI OCR accuracy, schedules vehicle inspections, approves permits, and routes for signing.	BPLO Operations Dashboard
IT System Administrator	Governs user accounts (RBAC), monitors server telemetry, configures municipal ordinances and fees, and oversees weekly database backups.	IT System Administration Console

2. SYSTEM REQUIREMENTS AND ACCESS

2.1 Hardware and Device Requirements

G-TRAMS is engineered with a mobile-first, responsive Progressive Web App (PWA) architecture. Users can access the platform using any modern internet-connected computing device:

- **For Operators & Enforcers (Mobile):** Any Android smartphone (Android 8.0 or higher) or Apple iPhone (iOS 13.0 or higher) with a working camera for QR code scanning and document photography.
- **For Municipal Staff & Administrators (Desktop/Laptop):** Windows 10/11, macOS, or Linux workstation with at least 4 GB RAM and a display resolution of 1280×720 or higher.
- **Printing Hardware:** Standard thermal receipt printer or A4 laser/inkjet printer for printing MTOP Certificates and Daily Remittance Sheets.

2.2 Web Browser and Software Requirements

SUPPORTED BROWSER	MINIMUM VERSION	RECOMMENDED SETTINGS
Google Chrome	Version 90 or higher	JavaScript enabled, Camera permission allowed
Microsoft Edge	Version 90 or higher	Chromium-based, pop-ups enabled for receipts
Mozilla Firefox	Version 88 or higher	Standard tracking protection enabled
Apple Safari	Version 14 or higher	iOS / iPadOS / macOS compatible

2.3 Internet Connectivity

G-TRAMS requires an active internet connection (Mobile 4G/5G, Broadband, or Wi-Fi) with a minimum recommended speed of **1 Mbps** for operators and **5 Mbps** for municipal office workstations handling multiple document uploads.

2.4 System Access URL

> **Official Production Access Link**
>
> Users and municipal personnel can access the live system at:
> **System URL:** https://g-trams-official.vercel.app/

3. GETTING STARTED

3.1 How to Access the System

1. Open your preferred web browser on your computer or mobile device.
2. Enter the official address: `https://g-trams-official.vercel.app` into the URL bar and press **Enter**.
3. The public landing page will display municipal announcements, quick links, and options to Log In or Register.

3.2 How to Register a New Account

Step 1: From the top navigation or login page, click the **Register** button.

Step 2: Enter your **Full Legal Name**, active **11-digit Mobile Number** (e.g. 09123456789), and valid **Email Address**.

Step 3: Select your **Barangay of Residence** within Gasan and your associated **TODA Group**.

Step 4: Create a secure password (minimum 8 characters with at least one letter and one number) and confirm it.

Step 5: Click **Create Account**. The system will prompt you for an SMS/Email One-Time Password (OTP) verification code.

Step 6: Enter the 6-digit OTP code received on your phone to activate your account.

Figure 2. G-TRAMS Account Registration Portal

3.3 How to Log In and Log Out

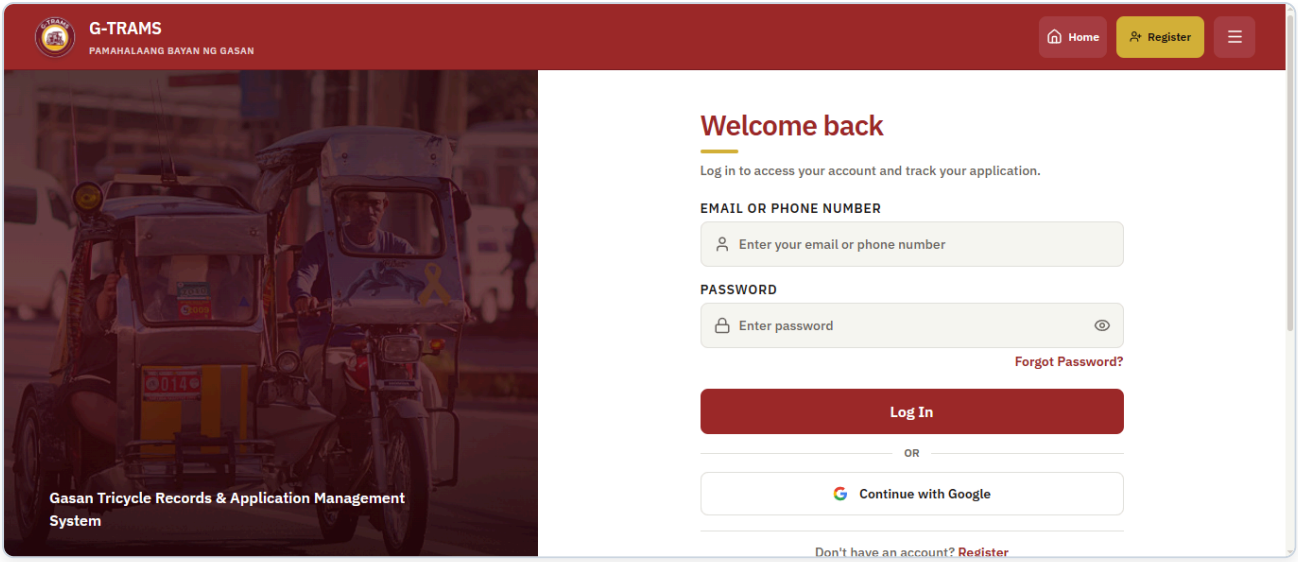


Figure 3. Official G-TRAMS Login Portal

Logging In:

1. Navigate to `/login`.
2. Enter your registered **Email Address** or **Mobile Phone Number** in the first field.
3. Enter your confidential **Password** in the password box.
4. Click the **Log In** button. The system will authenticate your credentials and automatically route you to your designated portal based on your assigned role.

Logging Out:

1. Locate the sidebar navigation on desktop or the profile menu on mobile.
2. Click the **Log Out** button with the red exit icon.
3. Your session token will be immediately cleared from local storage to prevent unauthorized access.

3.4 Forgot Password & Account Recovery

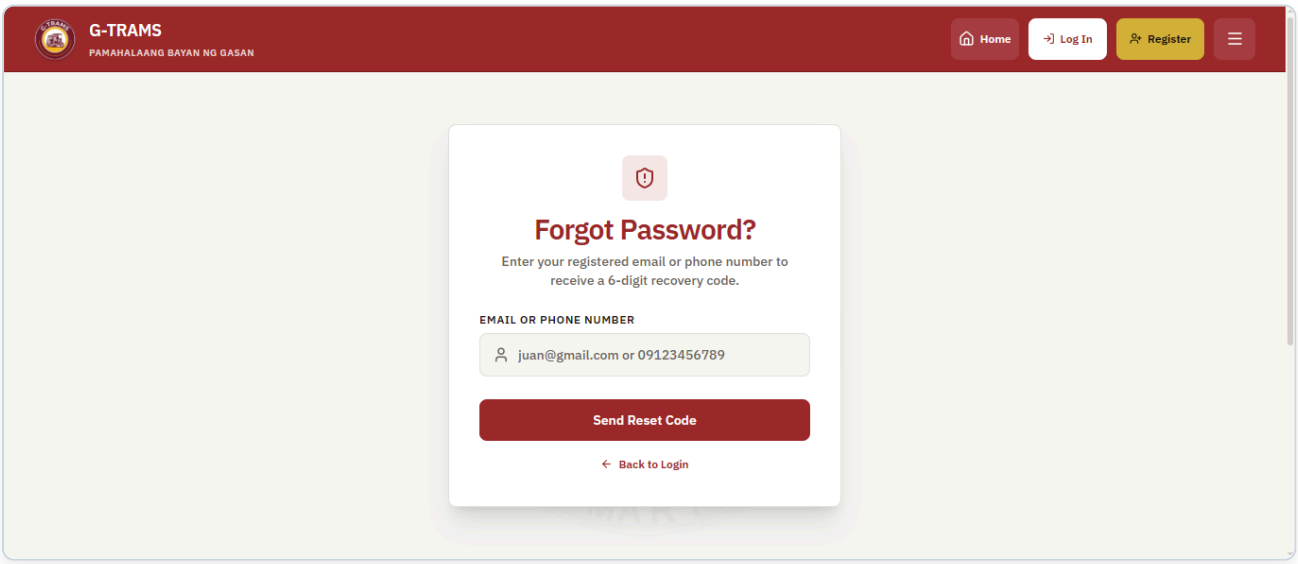


Figure 4. G-TRAMS Self-Service Password Recovery Portal

If you forget your password, click the **Forgot Password?** link on the login page. Enter your registered email or phone number. The system will dispatch a secure 6-digit recovery OTP code. Once verified, you will be prompted to establish a fresh, secure password.

4. STEP-BY-STEP USER INSTRUCTIONS

4.1 Citizen / Tricycle Operator Portal

The Operator Portal enables tricycle owners to manage vehicle registrations, apply for new MTOP franchises, submit required credentials, track inspection progress, and display digital certificates.

Feature 4.1.1: Navigating the Operator Garage & Unit Capacity

Purpose: View all registered tricycles under the operator's name and monitor remaining franchise slot capacity (maximum 2 units per operator as mandated by municipal ordinance).

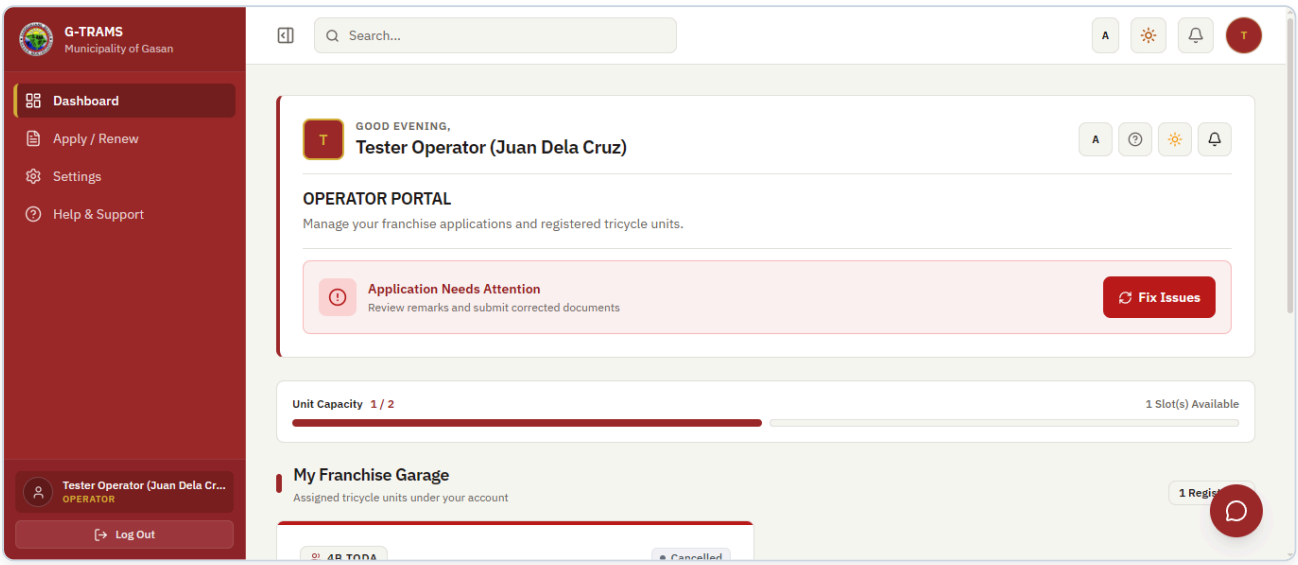


Figure 5. Operator Dashboard and Franchise Garage Overview

1. Log in with your Operator account. You will arrive at the **Operator Dashboard**.
2. Review the **Unit Capacity** widget at the top (e.g., "1 / 2 Slots Available").
3. In the **My Franchise Garage** section, view all registered units with their Plate Number, Motorcycle Model, Body Number, Route Zone, and Current Status (e.g., Active, For Payment, Needs Attention).
4. Click **View Details** on any vehicle card to see full application history and remarks.

Expected Result: The operator gains immediate visibility into vehicle legal statuses and active municipal permits.

Feature 4.1.2: Filing a New Franchise Application

Purpose: Submit an application for an unfranchised tricycle unit with digital document scanning.

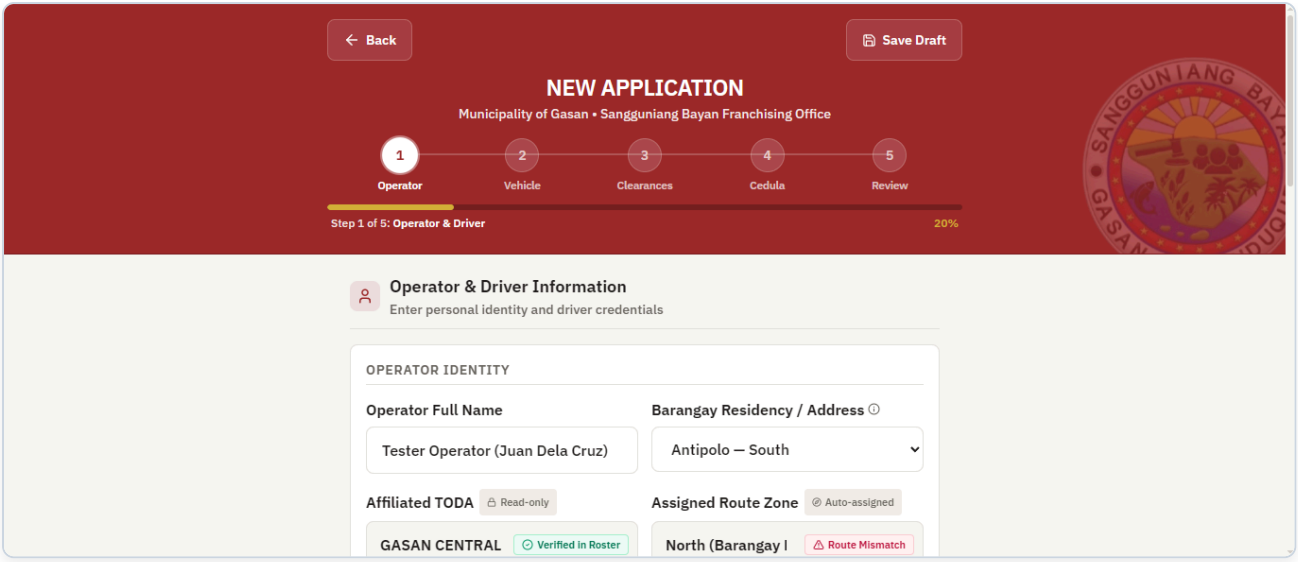


Figure 6. G-TRAMS Multi-Step Franchise Application Form

1. From the sidebar navigation, click **Apply / Renew**.
2. Select the application type: **New Franchise Application**.
3. Fill out the vehicle technical specifications: **Plate Number**, **Engine/Motor Number**, **Chassis Number**, **Make/Brand** (e.g., Honda, Kawasaki), and **Year Model**.
4. Select your designated **TODA Association** (e.g., Gasan Central TODA, Bahi TODA).
5. Upload the required supporting credentials (see Feature 4.1.3 below).
6. Review the municipal declaration statement and click **Submit Application**.

Expected Result: The application is queued with status **"Pending Review"** and routed directly to the BPLO Franchising Officer queue.

Feature 4.1.3: Document Upload with AI OCR Validation

Purpose: Upload clear photos of mandatory legal documents for automated pre-verification.

REQUIRED DOCUMENT	ACCEPTABLE FORMATS	SPECIAL REQUIREMENT
LTO Official Receipt (OR)	JPG, PNG, PDF (Max 10MB)	Current year validity, legible plate number
LTO Certificate of Registration (CR)	JPG, PNG, PDF (Max 10MB)	Registered owner must match operator legal name
Community Tax Certificate (Cedula)	JPG, PNG, PDF (Max 10MB)	Issued by Municipality of Gasan for current year
TODA Clearance / Endorsement	JPG, PNG, PDF (Max 10MB)	Signed by the accredited TODA President
Professional Driver's License	JPG, PNG, PDF (Max 10MB)	Valid Restriction Code 1 or DL Code A/A1

> **AI OCR Compliance Rule**
>
> Ensure all photos are taken in well-lit conditions with no glare or blurry corners. The system's built-in optical scanner automatically extracts plate numbers and motor numbers to verify document authenticity.

4.2 TODA President Portal

TODA Presidents oversee association membership, ensure route compliance, register new drivers under their umbrella, and endorse legitimate applicants to the LGU.

Feature 4.2.1: TODA Roster Management & Member Directory

Purpose: Maintain an official, audited directory of all operators and drivers operating within the association's authorized route.

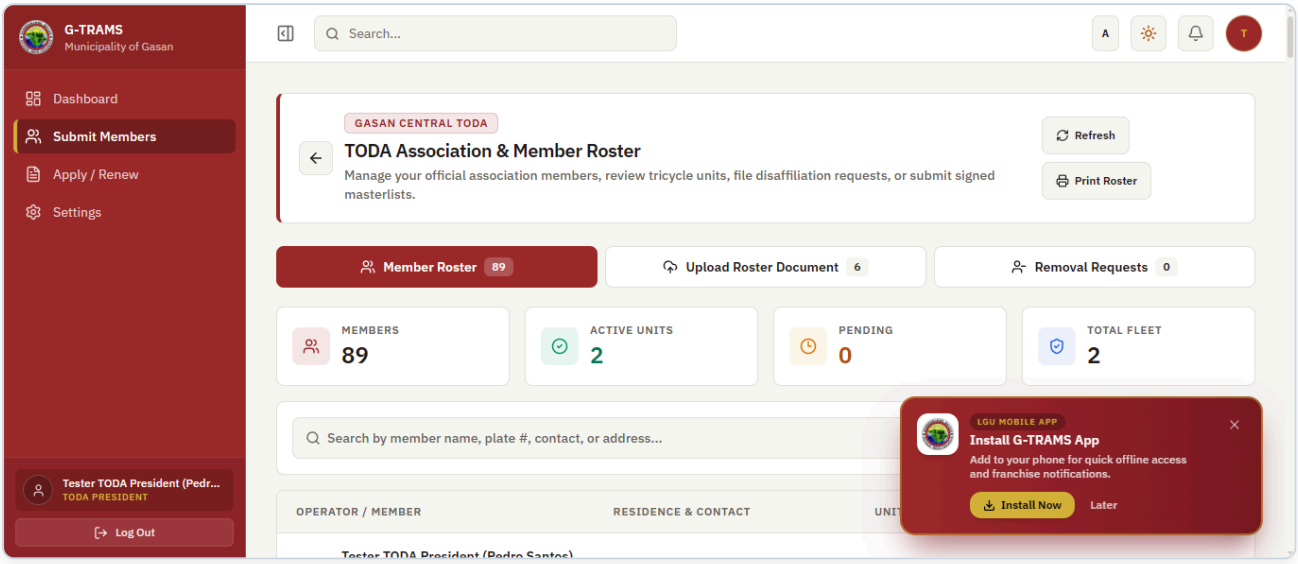


Figure 7. TODA President Member Submission and Roster Management Console

1. Log in with your TODA President account and click **Submit Members** from the sidebar.
2. The console displays your association's total registered members, active tricycle units, and pending clearance requests.
3. To add an individual member, click the **+ Add Individual Member** button.
4. Input the member's **Full Legal Name**, **Contact Number**, **Plate Number**, and **Tricycle Body Number**.
5. Click **Save to Official Roster**.

Expected Result: The member is immediately entered into the municipal TODA database and verified against duplicate body number assignments.

4.3 Traffic Enforcement Officer Portal (Mobile Web)

Enforcers from the Gasan Municipal Police Station (MPS), Traffic Management Unit, and POSO use the mobile web portal for rapid, on-road vehicle compliance auditing.

Feature 4.3.1: Live Mobile QR Code Scanning (Dual-Lookup)

Purpose: Instantly verify franchise legality by scanning either the Operator's Physical ID Card OR the Tricycle's MTOP Vehicle Certificate.

1. Open the G-TRAMS portal on your mobile smartphone browser.
2. Navigate to the QR scanner or point the phone's native camera at the vehicle's MTOP sticker or driver's ID QR code.
3. The system invokes the **Dual-Lookup Engine** to retrieve real-time compliance data.
4. The verification screen instantly displays:
   - **Franchise Status Badge:** Green (ACTIVE), Amber (FOR PAYMENT), or Red (EXPIRED / CANCELLED).
   - **Assigned Body Number & Plate Number.**
   - **Authorized TODA Association & Route Zone.**
   - **Permit Expiration Date.**

Expected Result: The enforcer can immediately identify whether a unit is legitimately franchised or operating illegally ("colorum").

Feature 4.3.2: Field Citation Issuance

Purpose: Log traffic violations (e.g., Overcharging, Out of Route, No Valid Permit) directly into the municipal database.

1. From the verified unit screen, click **Issue Citation Ticket**.
2. Select the specific violation from the Municipal Traffic Ordinance dropdown list.
3. Enter the exact location in Gasan (e.g., Poblacion Market, National Highway).
4. Upload optional photographic proof.
5. Click **Submit Citation**.

Expected Result: An automated SMS warning is immediately dispatched to the operator's registered mobile number, and the fine is added to the Cashier settlement ledger.

4.4 Municipal Cashier & Treasury Portal

The Municipal Cashier Terminal handles fee collection, official receipt logging, and revenue accounting in compliance with Commission on Audit (COA) standards.

Feature 4.4.1: Fee Settlement & Official Receipt (OR) Issuance

Purpose: Collect annual franchise fees or citation penalties and record official government receipts.

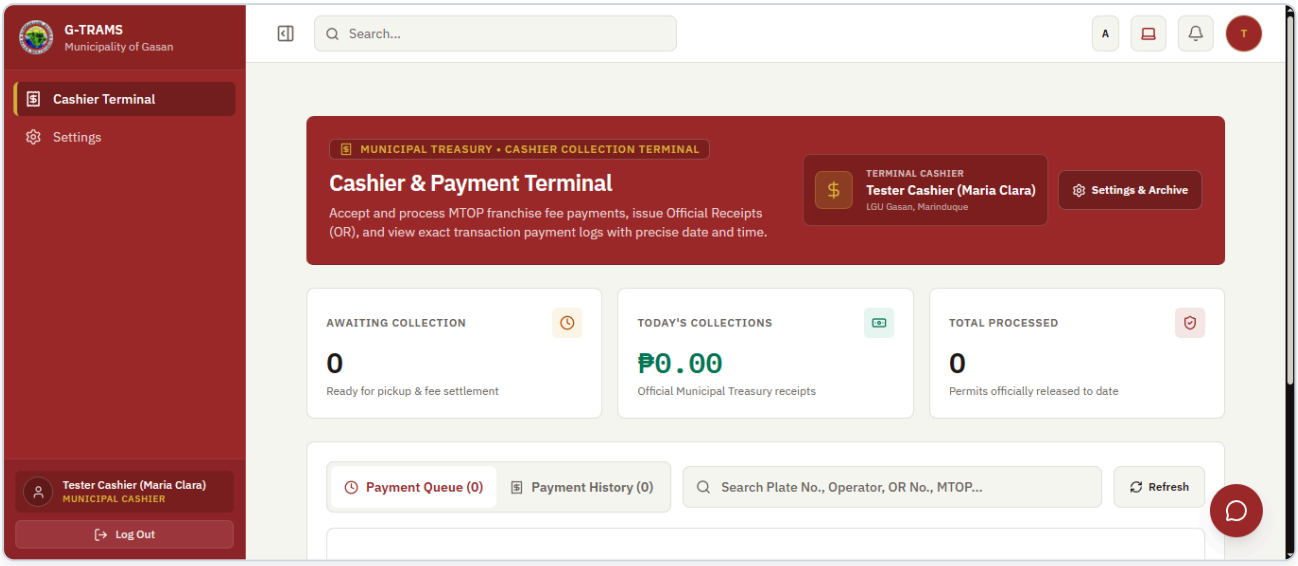


Figure 8. Municipal Treasury Cashier Collection Terminal

1. Log in with your Cashier account to reach the **Cashier Collection Terminal**.
2. In the **Awaiting Collection** queue, locate the applicant by typing their Plate Number, Operator Name, or MTOP Application ID in the search bar.
3. Click the **Process Payment** button corresponding to the record.
4. The assessment modal displays the fee breakdown (Franchise Fee, Sticker Fee, Inspection Fee).
5. Select payment mode: **Over-the-Counter Cash** or **Digital GCash**.
6. Input the official government **Official Receipt (OR) Number** and amount tendered.
7. Click **Confirm & Issue Official Receipt**.

Expected Result: The payment is recorded with precise timestamp, the application advances to **"For Signing"**, and an official printable receipt voucher is generated.

Feature 4.4.2: Daily Treasury Collection & Remittance Sheet

Purpose: Generate an audited daily remittance statement with treasurer signature blocks and 1-click CSV spreadsheet export.

1. From the top right of the Cashier Terminal, click the **Remittance Sheet** button.
2. The system aggregates all transactions completed during the business day, categorized into Cash Collections and Digital Collections.
3. Click **Print Remittance Sheet** to open the COA-compliant printable format with formal signature lines for the **Cashier / Disbursing Officer** and the **Municipal Treasurer**.
4. Click **Export to CSV** to immediately download the raw spreadsheet record: `GTRAMS_Treasury_Daily_Remittance_YYYY-MM-DD.csv`.

Expected Result: Complete financial transparency with zero discrepancy between cash drawers and digital records.

4.5 BPLO Franchising & Licensing Staff Portal

The Business Permits and Licensing Office (BPLO) manages the core approval pipeline, performs document scrutiny, schedules physical unit inspections, and issues MTOP certificates.

Feature 4.5.1: BPLO Operations Dashboard & Queue Counters

Purpose: Monitor application queues across all 5 municipal processing stages.

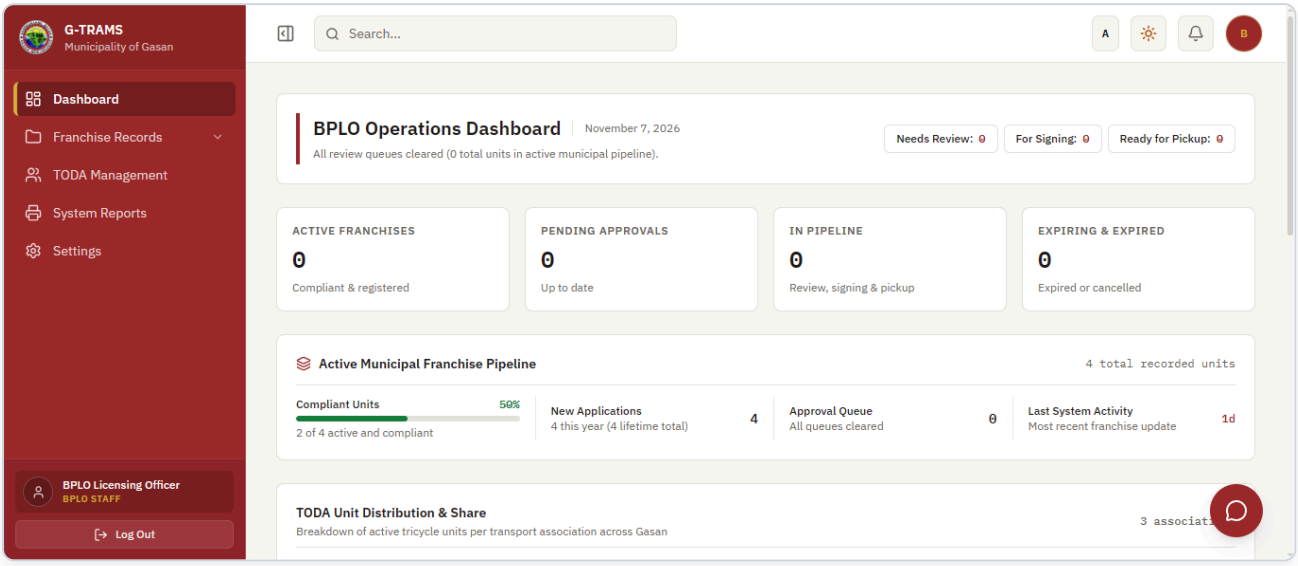


Figure 9. BPLO Operations Dashboard and Real-Time Pipeline Counters

The dashboard categorizes pending workloads into three immediate actionable queues: **Needs Review**, **For Signing**, and **Ready for Pickup**.

Feature 4.5.2: In-Depth Document Review & Approval Pipeline

Purpose: Inspect uploaded documents against OCR extracted values and approve applications.

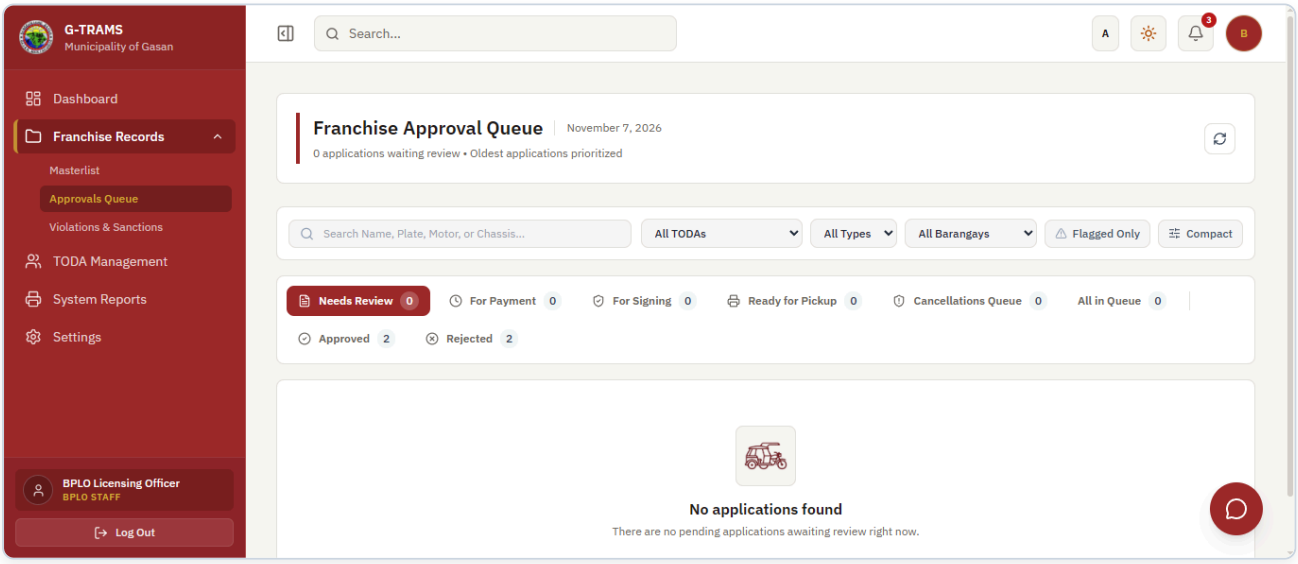


Figure 10. Franchise Approval and Document Inspection Queue

1. From the sidebar, click **Franchise Approval**.
2. Click **Review** on any pending applicant.
3. Side-by-side comparison opens showing the uploaded OR/CR images and the system's OCR match score.

4. Verify that the CR Owner matches the Operator name and that all numbers match.
5. If compliant, select **Approve & Route to Payment**. If issues are found, click **Request Correction** and specify the reason.

Expected Result: Operators receive instant SMS and email notifications informing them of their application approval or requested document fixes.

Feature 4.5.3: Franchise Masterlist & Archival Search

Purpose: Search, filter, and audit all historical and active tricycle records in the municipality.

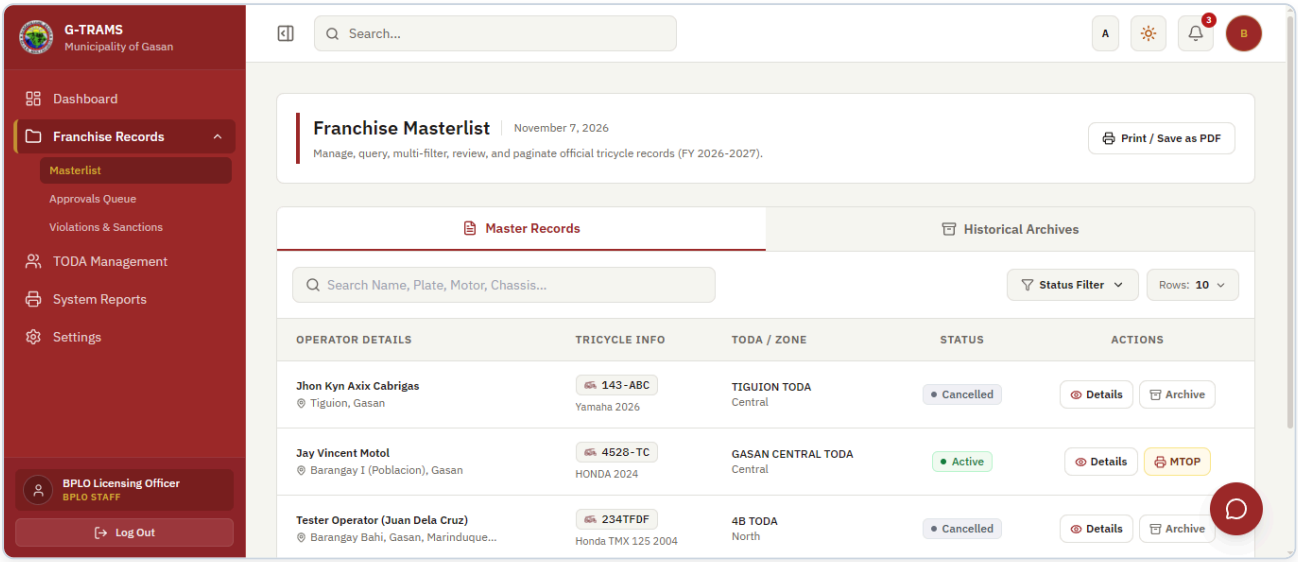


Figure 11. Complete Municipal Franchise Masterlist Directory

4.6 IT System Administrator Portal

The System Administrator console provides IT governance, infrastructure telemetry, user role assignment, and disaster recovery data protection.

Feature 4.6.1: IT Telemetry & Infrastructure Health

Purpose: Monitor live server health, database round-trip latency, and system security events.

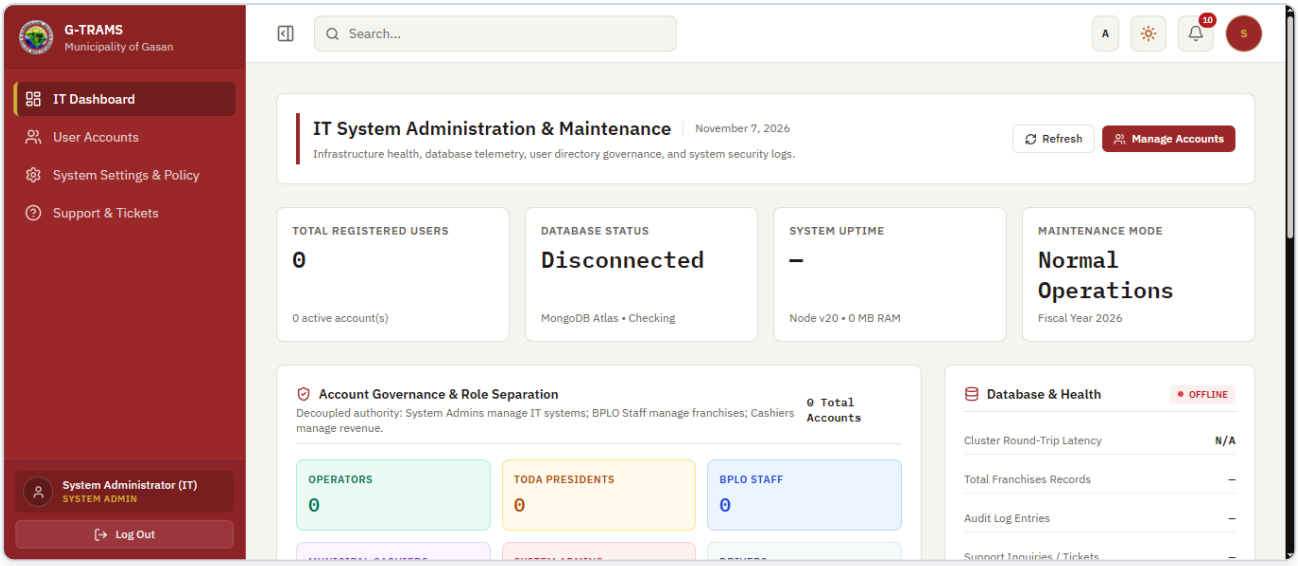


Figure 12. IT System Administration and Telemetry Dashboard

Feature 4.6.2: User Accounts Directory & Role Separation

Purpose: Manage staff user permissions and enforce Role-Based Access Control (RBAC).

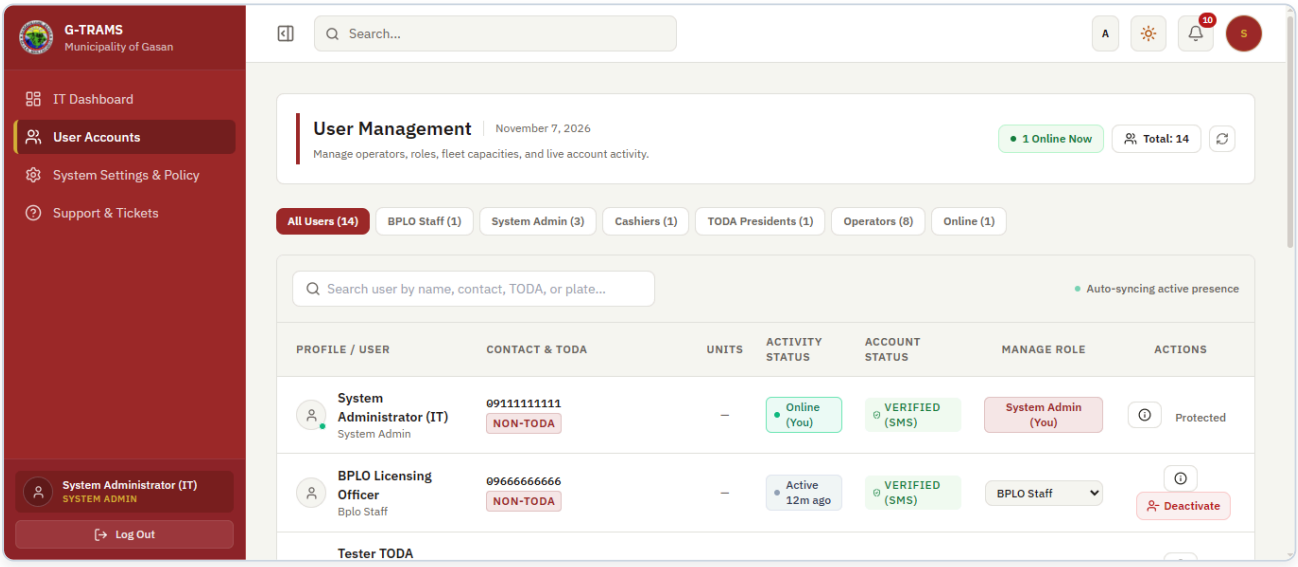


Figure 13. System User Accounts Directory and RBAC Governance

1. Click **User Accounts** from the admin sidebar.
2. Filter by role: Operator, TODA President, BPLO Staff, Cashier, or System Admin.
3. To change a user's role or deactivate an account, click the corresponding action button.

Feature 4.6.3: Automated Weekly Database Backup & Disaster Recovery (DRP)

Purpose: Safeguard municipal data with automated weekly cron snapshots and instant offline downloads.

1. Navigate to **System Settings** → **Tab 5: Database Backup & Disaster Recovery**.
2. Review the automated backup status: runs automatically **every Sunday at 12:00 Midnight (PST)** with an 8-week rolling retention policy.
3. To take an immediate backup before server maintenance, click **Backup Database Now**.
4. To archive the database directly to a local computer or external USB drive, click **Direct Offline Download**. The system streams a compressed Gzip archive (`.json.gz`) containing all 13 database collections.
5. To restore data after hardware replacement, click **Upload & Restore**, choose the archive, and type `RESTORE` to confirm.

> **Data Privacy Act (RA 10173) Backup Sanitization**
>
> During backup generation, all temporary authentication tokens and active OTP codes are stripped from the archive, while passwords remain protected by one-way bcrypt encryption.

5. ERROR HANDLING AND TROUBLESHOOTING

This section provides actionable solutions to common operational challenges encountered by users and municipal staff.

PROBLEM / ERROR	LIKELY CAUSE	RECOMMENDED CORRECTIVE ACTION
"Invalid email or password" error on login	Typographical error in email/phone or incorrect password.	Check Caps Lock. Use the "Show Password" eye icon to verify spelling. If forgotten, use the "Forgot Password?" link to reset via SMS OTP.
OTP SMS not received within 2 minutes	Network carrier signal delay or incorrect mobile number format.	Ensure number starts with '09' and is 11 digits long. Click "Resend OTP Code". Verify phone has cellular signal reception.
"Document rejected / Low match score"	Uploaded image is blurry, has light reflections, or corners are cropped out.	Retake the photograph in good lighting on a dark, flat surface. Ensure all text (especially Plate No. and Motor No.) is sharp and clear.
"Body Number Already In Use" conflict	Another tricycle in the same TODA association already holds that number.	The system strictly prevents duplicates. The TODA President or BPLO staff must assign the next available unique body number.
"Access Denied / 403 Forbidden"	Attempting to access a portal outside your assigned role permissions.	Log out and log in with the correct account. For staff role upgrades, contact the IT System Administrator.
Page fails to load or displays spinner	Intermittent internet connection or stale browser cache.	Perform a hard refresh (`Ctrl + F5` on Windows or `Cmd + Shift + R` on Mac). Clear browser cookies if problem persists.

6. FREQUENTLY ASKED QUESTIONS (FAQS)

Q1: Ilang tricycle ang pinapayagang ma-rehistro ng isang operator sa ilalim ng G-TRAMS?

Sagot: Ayon sa Municipal Ordinance ng Bayan ng Gasan, ang bawat rehistradong operator ay pinapayagan lamang magkaroon ng hanggang **dalawang (2) prangkisa** upang magkaroon ng pantay na oportunidad ang lahat ng mga mamamayan.

Q2: Paano ko malalaman kung approved na ang aking franchise application?

Sagot: Makakatanggap ka ng opisyal na **SMS notification** sa iyong rehistradong cellphone number sa bawat pagbabago ng estado ng iyong aplikasyon. Maaari mo ring tingnan ang iyong Operator Dashboard anumang oras.

Q3: Pwede ba akong magbayad sa pamamagitan ng GCash sa halip na magdala ng cash?

Sagot: Opo. Ang Municipal Treasury Cashier Terminal ay tumatanggap ng parehong Over-the-Counter Cash at Digital GCash transfers. Ipakita lamang ang inyong GCash transaction reference number sa cashier para sa issuance ng inyong Official Receipt.

Q4: Ano ang dapat gawin ng enforcer kung walang internet signal sa daan habang nag-i-scan?

Sagot: Ang G-TRAMS Mobile PWA ay may offline caching capability. Kung pansamantalang walang signal, basahin ang nakalimbag na impormasyon sa physical MTOP Certificate at i-recheck ang QR code sa sandaling magkaroon muli ng mobile data connectivity.

Q5: Ano ang pagkakaiba ng QR Code sa Operator ID kumpara sa QR Code sa MTOP Certificate?

Sagot: Ang system ay may **Dual-Lookup Engine**. Ang QR code sa Operator ID ay nagpapatunay ng pagkakakilanlan ng drayber/operator, habang ang QR sa MTOP Certificate ay nagpapatunay ng legalidad ng mismong sasakyan. Pareho itong gagana kapag ini-scan ng traffic enforcers.

Q6: Paano kung lumipat ng ibang TODA ang isang miyembro?

Sagot: Kailangang maghain ang dating TODA President ng **Member Removal Request** kalakip ang clearance, at mag-apply ang operator ng TODA Transfer sa BPLO portal.

Q7: Ligtas ba ang aking mga personal na dokumento sa system?

Sagot: Opo. Ang lahat ng mga dokumento at personal na impormasyon ay naka-imbak sa mga secure cloud servers na sumusunod sa mga pamantayan ng Data Privacy Act of 2012 (RA 10173).

7. SECURITY AND USER REMINDERS

To ensure the integrity, confidentiality, and reliability of G-TRAMS, all users and municipal personnel must strictly observe the following security protocols:

1. Protection of Account Credentials & Passwords

Never share your login credentials, passwords, or SMS OTP codes with anyone. Municipal employees will never ask for your password. Ensure your password is at least 8 characters with numbers and special symbols.

2. Mandatory Session Logout on Shared Workstations

Always click the **Log Out** button before leaving any computer terminal, especially in municipal offices, internet cafes, or shared family devices. Closing the browser window without logging out may leave your session vulnerable.

3. Compliance with Data Privacy Act of 2012 (RA 10173)

All information uploaded to G-TRAMS—including photos, contact numbers, and vehicle credentials—is classified as official regulatory data. Republic Act 10173 strictly penalizes unauthorized disclosure, copying, or dissemination of personal data.

4. Lawful System Usage

This system is intended exclusively for authorized tricycle regulatory management in the Municipality of Gasan. Falsification of uploaded documents, submission of fraudulent receipts, or attempted unauthorized access constitutes a criminal offense under Philippine cybercrime and local regulatory laws.

8. SUPPORT AND CONTACT INFORMATION

If you experience technical errors, account lockouts, or have operational inquiries regarding franchise applications, contact the appropriate municipal office or development representative:

OFFICE / CHANNEL	PHYSICAL LOCATION	CONTACT INFORMATION
Business Permits & Licensing Office (BPLO)	Ground Floor, Municipal Hall, Barangay I (Poblacion), Gasan, Marinduque	Phone: (042) 342-1234 Email: `bplo@gasan.ph` Office Hours: Mon–Fri, 8:00 AM – 5:00 PM
Municipal Treasury Office	Ground Floor, Municipal Hall, Gasan, Marinduque	Phone: (042) 342-1235 Email: `treasury@gasan.ph` Office Hours: Mon–Fri, 8:00 AM – 4:30 PM
G-TRAMS Technical Support Team	College of Information and Computing Sciences, Marinduque State University	Group Lead: Jhon Kyn Axix H. Cabrigas Frontend Support: Justine S. Lachica Email: `support@g-trams-official.vercel.app`

MUNICIPALITY OF GASAN • PROVINCE OF MARINDUQUE

Empowering Public Transport Transparency through Modern Digital Governance

G-TRAMS System User's Manual • Version 2.4.0 • Academic Year 2026–2027